Session 1: Introduction to Statistics in Data Science

Assignment Solutions & Data Classification Guide

1. Daily App Actions & Data Classification

Analyzing 5 common user actions on popular mobile applications and classifying the generated data type:

App	User Action	Data Generated	Data Type
Instagram	Liking a post / reel	Total like count (e.g., 1,420 likes)	**Numerical** (Discrete)
Zomato	Selecting cuisine category filter	Cuisine chosen (e.g., "North Indian", "Chinese")	**Categorical** (Nominal)
Flipkart	Rating a product after delivery	Star rating score (e.g., 4 out of 5 stars)	**Categorical** (Ordinal)
Zomato	Tracking delivery status	Estimated arrival time (e.g., 28.5 minutes)	**Numerical** (Continuous)
Instagram	Posting a story update	Filter/Effect used (e.g., "Clarendon", "Normal")	**Categorical** (Nominal)

2. Music App Genre Data Classification

Dataset: `['Pop', 'Rock', 'Jazz', 'Hip-Hop', 'Pop', 'Rock', 'Jazz', 'Pop']`

Classification: Categorical (Nominal) Data

Reasoning: This dataset represents qualitative labels (music genres) that describe distinct categories rather than numerical measurements. It is specifically *nominal* because there is no natural order, ranking, or quantitative distance between music styles like 'Pop' and 'Jazz'.

3. Improving App Features via Statistics: Zomato Restaurant Ratings

Feature: Restaurant Rating & Review System

Instead of showing a simple arithmetic mean rating (which can be heavily distorted by fake 1-star reviews or old historical data), Zomato can use **Weighted Bayesian Averages** and **Time-Decay Variance Analysis**. By weighing recent ratings more heavily and calculating the variance of scores, the app can display a *"Confidence-Adjusted Rating Score"* alongside a consistency badge (e.g., "95% consistent quality over last 30 days"), giving users a much more reliable metric.

4. Statistical Terms for Spotify Recommendation Engines

3 essential statistical concepts applied to analyze user listening habits for smart playlist generation:

1. Mode

Identifies the most frequently played music genre or artist in a user's history to establish their primary preference.

Example: Finding that a user listened to "Lo-Fi" 42 times this week (the mode) to auto-generate a chill playlist.

2. Median

Determines the middle song duration or tempo (BPM), remaining robust against extreme outliers like 30-minute podcasts.

Example: Recommending songs around the median tempo of 120 BPM while ignoring one 3-hour ambient track.

3. Variance

Measures the diversity or spread in a user's listening habits to decide whether to recommend tight or varied songs.

Example: High variance in audio features triggers an "Eclectic Mix", whereas low variance generates a focused genre radio.